

Bussiness Bean-

```
package com.accenture.lkm.business.bean;

import java.util.Date;

import javax.persistence.Temporal;
import javax.persistence.TemporalType;

public class CustomerBean {

    private int customerId;
    private String customerName;
    private Date billingDate;
    private Double bill;
    private String customerType;
    public int getCustomerId() {
        return customerId;
    }
    public void setCustomerId(int customerId) {
        this.customerId = customerId;
    }
    public String getCustomerName() {
        return customerName;
    }
    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }
    public Date getBillingDate() {
        return billingDate;
    }
    public void setBillingDate(Date billingDate) {
        this.billingDate = billingDate;
    }
    public Double getBill() {
        return bill;
    }
    public void setBill(Double bill) {
        this.bill = bill;
    }
    public String getCustomerType() {
        return customerType;
    }
    public void setCustomerType(String customerType) {
        this.customerType = customerType;
    }
    @Override
    public String toString() {
        return "CustomerBean [customerId=" + customerId + ",
customerName=" + customerName + ", billingDate="
            + billingDate + ", bill=" + bill + ",
customerType=" + customerType + "]\n";
    }
}
```

}


```
package com.accenture.lkm.business.bean;
```

```
import java.util.Date;
```

```
public class DateRangeBean {
```

```
    private Date fromDate;
```

```
    private Date toDate;
```

```
    public Date getFromDate() {  
        return fromDate;
```

```
    }
```

```
    public void setFromDate(Date fromDate) {  
        this.fromDate = fromDate;
```

```
    }
```

```
    public Date getToDate() {  
        return toDate;
```

```
    }
```

```
    public void setToDate(Date toDate) {  
        this.toDate = toDate;
```

```
    }
```

```
}
```

```
-----  
-----
```

```
package com.accenture.lkm.business.bean;
```

```
public class CustomerTypeBean {
```

```
    private String customerType;
```

```
    private Double minimumBill;
```

```
    private Double maximumBill;
```

```
    public String getCustomerType() {  
        return customerType;
```

```
    }
```

```
    public void setCustomerType(String customerType) {  
        this.customerType = customerType;
```

```
    }
```

```
    public Double getMinimumBill() {  
        return minimumBill;
```

```
    }
```

```
    public void setMinimumBill(Double minimumBill) {  
        this.minimumBill = minimumBill;
```

```
    }
```

```
    public Double getMaximumBill() {  
        return maximumBill;
```

```
    }
```

```
    public void setMaximumBill(Double maximumBill) {  
        this.maximumBill = maximumBill;
```

```
    }
```

```
}
```

```
=====
```

```
DAO-
```

```
package com.accenture.lkm.dao;
```

```
import java.util.Date;
```

```

import java.util.List;

import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.RepositoryDefinition;
import org.springframework.transaction.annotation.Transactional;

import com.accenture.lkm.entity.CustomerEntity;

@RepositoryDefinition(idClass = Integer.class, domainClass =
CustomerEntity.class)
@Transactional(value = "txManager")
// name of TransactionManger can be any thing
public interface CustomerDAO {
    @Query(name = "GetCustomerDetailsQuery")
    List<CustomerEntity> getCustomerDetails(Date startDate, Date
endDate);

    //@Query(name = "DummyQuery")
    //List<CustomerEntity> getCustomerDetailsWithin(Date startDate,
Date endDate);
}
-----
package com.accenture.lkm.dao;

import java.util.ArrayList;
import java.util.Date;
import java.util.List;

import javax.persistence.EntityManager;
import javax.persistence.PersistenceContext;
import javax.persistence.Query;

import org.springframework.beans.BeanUtils;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;

import com.accenture.lkm.business.bean.CustomerBean;
import com.accenture.lkm.business.bean.CustomerTypeBean;
import com.accenture.lkm.entity.CustomerEntity;
import com.accenture.lkm.entity.CustomerTypeEntity;

@Repository
@Transactional(value = "txManager")
public class CustomerDAOWrapper {

    @Autowired
    private CustomerDAO customerDAO;

    @Autowired
    private CustomerTypeDAO customerTypeDAO;

    @PersistenceContext
    private EntityManager entityManager;

    public List<CustomerBean> getCustomerDetails(Date startDate, Date
endDate) throws Exception{
        List<CustomerBean> beanList = null;

```

```

        try {
            List<CustomerEntity> entityList =
customerDAO.getCustomerDetails(startDate, endDate);

            if(entityList!=null) {
                beanList = new ArrayList<CustomerBean>();

                for(CustomerEntity entity : entityList) {
                    CustomerBean bean = convertEntityToBean(entity);
                    beanList.add(bean);
                }

            }
        }catch(Exception e) {
            System.out.println(e.getMessage());
        }
        return beanList;
    }

    public List<String> getCustomerTypes() throws Exception{
        List<String> list = null;
        try{
            list = customerTypeDAO.getCustomerTypes();
        }catch(Exception e) {
            System.out.println(e.getMessage());
        }
        return list;
    }

    public CustomerTypeBean getCustomerTypeDetails(String customerType)
throws Exception{
        CustomerTypeBean bean = null;
        try {
            Query query = entityManager.createQuery("select c from
CustomerTypeEntity c where c.customerType=?1");
            query.setParameter(1, customerType);
            CustomerTypeEntity customerTypeEntity=
(CustomerTypeEntity)query.getSingleResult();
            bean = convertTypeEntityToTypeBean(customerTypeEntity);
        }
        catch(Exception e) {
            System.out.println(e.getMessage());
        }
        return bean;
    }

    public Integer addCustomer(CustomerBean customerBean) throws
Exception{
        CustomerEntity entity = new CustomerEntity();
        Integer id = null;
        try {
            entity = convertBeanToEntity(customerBean);
            entityManager.persist(entity);
            id = entity.getCustomerId();
        }
        catch(Exception e) {
            System.out.println(e.getMessage());
        }
        return id;
    }

```

```

    }

    // Utility Methods.....
    public static CustomerBean convertEntityToBean(CustomerEntity
entity) {
        CustomerBean customerBean = new CustomerBean();
        BeanUtils.copyProperties(entity, customerBean);
        return customerBean;
    }
    public static CustomerEntity convertBeanToEntity(CustomerBean bean)
{
        CustomerEntity entity = new CustomerEntity();
        BeanUtils.copyProperties(bean, entity);
        return entity;
    }

    public static CustomerTypeBean
convertTypeEntityToTypeBean(CustomerTypeEntity entity) {
        CustomerTypeBean customerTypeBean = new CustomerTypeBean();
        BeanUtils.copyProperties(entity, customerTypeBean);
        return customerTypeBean;
    }
}

-----
package com.accenture.lkm.dao;

import java.util.List;

import org.springframework.data.jpa.repository.Query;
import org.springframework.data.repository.RepositoryDefinition;
import org.springframework.transaction.annotation.Transactional;

import com.accenture.lkm.entity.CustomerTypeEntity;

@RepositoryDefinition(idClass = String.class, domainClass =
CustomerTypeEntity.class)
@Transactional(value = "txManager")
public interface CustomerTypeDAO {

    @Query(name = "GetCustomerTypesQuery")
    List<String> getCustomerTypes();

}
-
=====
ENTITY-
package com.accenture.lkm.entity;

import java.util.Date;

import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;

```

```
import javax.persistence.Table;
import javax.persistence.Temporal;
import javax.persistence.TemporalType;
```

```
@Entity
@Table(name = "Customer")
public class CustomerEntity {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int customerId;
    private String customerName;
    @Temporal(TemporalType.DATE)
    private Date billingDate;
    private Double bill;
    private String customerType;
    public int getCustomerId() {
        return customerId;
    }
    public void setCustomerId(int customerId) {
        this.customerId = customerId;
    }
    public String getCustomerName() {
        return customerName;
    }
    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }
    public Date getBillingDate() {
        return billingDate;
    }
    public void setBillingDate(Date billingDate) {
        this.billingDate = billingDate;
    }
    public Double getBill() {
        return bill;
    }
    public void setBill(Double bill) {
        this.bill = bill;
    }
    public String getCustomerType() {
        return customerType;
    }
    public void setCustomerType(String customerType) {
        this.customerType = customerType;
    }
}
```

```
}
-----
-----
package com.accenture.lkm.entity;
```

```
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;
```

```
@Entity
@Table(name = "CustomerTypes")
```

```

public class CustomerTypeEntity {

    @Id
    private String customerType;
    private Double minimumBill;
    private Double maximumBill;
    public String getCustomerType() {
        return customerType;
    }
    public void setCustomerType(String customerType) {
        this.customerType = customerType;
    }
    public Double getMinimumBill() {
        return minimumBill;
    }
    public void setMinimumBill(Double minimumBill) {
        this.minimumBill = minimumBill;
    }
    public Double getMaximumBill() {
        return maximumBill;
    }
    public void setMaximumBill(Double maximumBill) {
        this.maximumBill = maximumBill;
    }
}

=====
=====
Resources-

-- Table Script goes here.
DROP DATABASE IF EXISTS springmvcmock;
create database springmvcmock;
use springmvcmock;
create table if not exists Customer(
    customerId int(10) auto_increment,
    customerName varchar(20),
    billingDate date,
    bill double(10,2),
    customerType varchar(3),
    PRIMARY KEY(customerId));

create table customerTypes(
    customerType varchar(3),
    minimumBill double(10,2),
    maximumBill double(10,2),
    PRIMARY KEY(customerType));

insert into Customer values
    (1001,'Jack','2018-01-12',6000,'P'),
    (1002,'Mark','2019-04-10',3000,'G');

insert into CustomerTypes values
    ('S',1001,2000),
    ('G',2001,5000),
    ('P',5001,10000);
=====
=====

```

```

Service-
package com.accenture.lkm.service;

import java.util.Date;
import java.util.List;
import java.util.Map;

import com.accenture.lkm.business.bean.CustomerBean;

public interface CustomerService {

    List<CustomerBean> getCustomerDetails(Date startDate, Date endDate)
throws Exception;
    Integer addCustomer(CustomerBean customerBean) throws Exception;
    Map<String, String> getCustomerTypes() throws Exception;

}
-----
package com.accenture.lkm.service;

import java.util.Date;
import java.util.HashMap;
import java.util.List;
import java.util.Map;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.accenture.lkm.business.bean.CustomerBean;
import com.accenture.lkm.business.bean.CustomerTypeBean;
import com.accenture.lkm.dao.CustomerDAOWrapper;

@Service
public class CustomerServiceImpl implements CustomerService{

    @Autowired
    private CustomerDAOWrapper customerDAOWrapper;

    @Override
    public List<CustomerBean> getCustomerDetails(Date startDate, Date
endDate) throws Exception{

        List<CustomerBean> beanList =
customerDAOWrapper.getCustomerDetails(startDate, endDate);
        if(beanList==null || beanList.isEmpty()) {
            System.out.println("No data found");
            throw new RuntimeException("No Customer found within
given date range");
        }
        else {
            return beanList;
        }
    }

    @Override
    public Integer addCustomer(CustomerBean customerBean) throws
Exception{
        CustomerTypeBean customerTypeBean = new CustomerTypeBean();

```



```

        customerTypeBean =
customerDAOWrapper.getCustomerTypeDetails(customerBean.getCustomerType())
;
        double maxBill = customerTypeBean.getMaximumBill();
        double minBill = customerTypeBean.getMinimumBill();

        if(customerBean.getBill()>=minBill &&
customerBean.getBill()<=maxBill) {
            return customerDAOWrapper.addCustomer(customerBean);
        }
        else {
            throw new RuntimeException("Bill is not in the given
range");
        }
    }

    @Override
    public Map<String, String> getCustomerTypes() throws Exception {
        List<String> customerTypesList =
customerDAOWrapper.getCustomerTypes();
        Map<String, String> map = new HashMap<String, String>();

        for(String type : customerTypesList) {

            switch(type) {
                case "P" : map.put(type, "Platinum");
                    break;
                case "G" : map.put(type, "Gold");
                    break;
                case "S" : map.put(type, "Silver");
                    break;
            }

        }

        return map;
    }
}

```

```

=====
=====

```

CONTROLLER-

```
package com.accenture.lkm.web.controller;
```

```
import java.util.Map;
```

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
import org.springframework.web.servlet.ModelAndView;
```

```
import com.accenture.lkm.business.bean.CustomerBean;
import com.accenture.lkm.service.CustomerService;
```

```
@Controller
```

```

public class CustomerController {

    @Autowired
    private CustomerService customerService;

    @RequestMapping("/addCustomerURL.html")
    public ModelAndView loadCustomerForm() {
        return new ModelAndView("CreateCustomer","customerBean", new
CustomerBean());
    }

    @RequestMapping(value="saveCustomer", method = RequestMethod.POST )
    public ModelAndView
saveCustomer(@ModelAttribute("customerBean")CustomerBean customerBean)
throws Exception{
        ModelAndView mv = new ModelAndView();
        Integer id = customerService.addCustomer(customerBean);
        mv.setViewName("CreateCustomerSuccess");
        mv.addObject("message", "Customer added successfully with id:"
+ id);
        return mv;
    }

    @ModelAttribute("customerTypesRadio")
    public Map<String, String> populateCustomerType() throws Exception{
        return customerService.getCustomerTypes();
    }

    //Error Handler:
    @ExceptionHandler(value=Exception.class)
    public ModelAndView handleAllExceptions(Exception exception){
        ModelAndView modelAndView = new ModelAndView();
        modelAndView.setViewName("GeneralizedExceptionHandlerPage");
        modelAndView.addObject("message", exception.getMessage());
        modelAndView.addObject("exception", exception);
        return modelAndView;
    }

}
-----
package com.accenture.lkm.web.controller;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.ModelAttribute;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
import org.springframework.web.servlet.ModelAndView;

import com.accenture.lkm.business.bean.CustomerBean;
import com.accenture.lkm.business.bean.DateRangeBean;
import com.accenture.lkm.service.CustomerService;

@Controller

```

```

public class ReportController {

    @Autowired
    private CustomerService customerService;

    @RequestMapping("/showCustomerDetails.html")
    public ModelAndView loadReportPage(){
        return new ModelAndView("CustomerDetails", "dateRangeBean",
new DateRangeBean());
    }

    @RequestMapping(value="/displayCustomerRecordsURL",
method=RequestMethod.GET)
    public ModelAndView
getCustomerRecords(@ModelAttribute("dateRangeBean") DateRangeBean
dateRangeBean) throws Exception{
        ModelAndView mv = new ModelAndView();
        List<CustomerBean> beanList =
customerService.getCustomerDetails(dateRangeBean.getFromDate(),
dateRangeBean.getToDate());
        mv.setViewName("CustomerDetails");
        mv.addObject("beanList", beanList);
        //System.out.println(beanList);
        return mv;
    }

    //Error Handler:
    @ExceptionHandler(value=Exception.class)
    public ModelAndView handleAllExceptions(Exception exception){
        ModelAndView modelAndView = new ModelAndView();

        modelAndView.setViewName("GeneralizedExceptionHandlerPage");
        modelAndView.addObject("message",
exception.getMessage());
        modelAndView.addObject("exception", exception);
        return modelAndView;
    }

}

=====
=====
=====
JSP Views-

CREATE CUSTOMER-

<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
    <%@taglib uri="http://www.springframework.org/tags/form"
prefix="form" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Customer Creation</title>
</head>
<body>
<center>

```

```

<h2>Add Customer Details</h2>
<form:form action="{pageContext.request.contextPath}/saveCustomer.html"
method="POST" modelAttribute="customerBean">
    <table border="2">
        <tr>
            <td>Customer Name</td>
            <td><form:input path="customerName"/></td>
        </tr>
        <tr>
            <td>Customer Type</td>
            <td><form:radiobuttons path="customerType"
items="{customerTypesRadio}"/></td>
        </tr>
        <tr>
            <td>Bill Amount</td>
            <td><form:input path="bill"/></td>
        </tr>
        <tr>
            <td>Billing Date[dd-MMM-yyyy]</td>
            <td><form:input path="billingDate"/></td>
        </tr>
    </table>
    <input type="submit" value="Register" /><br><br>
</form:form>

```

```

<a href="{pageContext.request.contextPath}/index.jsp">Home</a>

```

```

</center>
</body>
</html>

```

```

-----
CREATE CUSTOMER SUCESS-
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<%@ taglib uri="http://www.springframework.org/tags/form" prefix="form"%>
<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>Dummy Page</title>
<style>
.error {
    color: #ff0000;
    font-style: italic;
}
</style>
</head>
<body>
    <center>
        <h2>Add Customer Details Success</h2>
        ${message}
        <a
href="{pageContext.request.contextPath}/index.jsp">Home</a>
    </center>
</body>
</html>

```


CUSTOMER DETAILS-

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
```

```
<%@taglib uri="http://www.springframework.org/tags/form"
    prefix="form" %>
```

```
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c" %>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="ISO-8859-1">
```

```
<title>Customer Details</title>
```

```
</head>
```

```
<body>
```

```
<Center>
```

```
    <form:form
```

```
    action="${pageContext.request.contextPath}/displayCustomerRecordsURL.html
```

```
    " method="GET" modelAttribute="dateRangeBean">
```

```
        <table>
```

```
            <tr>
```

```
                                <td>From Date:</td><td><form:input
path="fromDate"/></td>
```

```
                                <td>To Date:</td><td><form:input
path="toDate"/></td>
```

```
            </tr>
```

```
        </table><br>
```

```
        <input type="submit" value="Fetch Details" /><br>
```

```
    </form:form>
```

```
    <a href="${pageContext.request.contextPath}/index.jsp">Home</a>
```

```
<c:if test="${not empty beanList }">
```

```
    <table border="2">
```

```
        <tr>
```

```
            <th>CustomerId</th><th>CustomerName</th><th>Bill</th><th>Billing
Date</th>
```

```
        </tr>
```

```
        <c:forEach var="list" items="${beanList}">
```

```
            <tr>
```

```
                <td>${list.customerId}</td><td>${list.customerName}</td><td>${list.
bill}</td><td>${list.billingDate}</td>
```

```
            </tr>
```

```
        </c:forEach>
```

```
    </table>
```

```
</c:if>
```

```
</Center>
```

```
</body>
```

```
</html>
```


GENERALIZEDEXCEPTIONANDELING

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<center>
<h2 style="color:red">Generalized Exception Handler Page</h2>
${message}<br>
${exception}
<br>
<br>
<a href="${pageContext.request.contextPath}/index.jsp">Home</a>
</center>
</body>
</html>
```

=====

INDEX-

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
    pageEncoding="ISO-8859-1"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<center>
    <h2>XYZ Customer Management System</h2>

    <a
href="${pageContext.request.contextPath}/addCustomerURL.html">Add
Customer</a> <br><br>
    <a
href="${pageContext.request.contextPath}/showCustomerDetails.html">Displa
y Customer Details</a>

</center>
</body>
</html>
```

=====

=====

===